Assignment 5 – Welcome to the World of Mario

Ling 201 – Fall 2026

(100 pts total)

Part 1. Presupposition Test *[difficulty level: ☆☆☆☆]*

For each pair of sentences below, determine **whether or not sentence (a) presupposes sentence (b)**, and **justify your answer using semantic reasoning** (truth conditions, etc.).

1. (a). Toad texted Toadette again. (b). Toad texted Toadette before.
   i. (a) entails (b), if Toad texted Toadette again, then he must have texted Toadette before. Hence, if (a) is **TRUE**, (b) must also be **TRUE**.
   ii. NOT-(a) entails (b), if Toad did not text Toadette again, then he must have texted Toadette before still. Hence, if NOT-(a) is **TRUE**, (b) must also be **TRUE**.
   iii. Since both (a) and NOT-(a) entail (b), (a) **PRESUPPOSES** (b)
2. (a). Baby Luigi did his homework. (b). Baby Luigi redid his homework.
   i. a) does not entail (b), if Baby Luigi did his homework, it is not necessarily true that he redid his homework, Hence, if (a) is **TRUE**, (b) is not necessarily **TRUE**.
   ii. Hence, (a) **does not PRESUPPOSE** (b)
3. (a). The world of Mario is fantastic.

   (b). Bowser knows that the world of Mario is fantastic.
   i. a) does not entail (b), if it is true that the world of Mario is fantastic, Bowser still does not necessarily know that. Hence, if (a) is **TRUE**, (b) is not necessarily **TRUE**.
   ii. Hence, (a) **does not PRESUPPOSE** (b)
4. (a). What Boo wanted was the crown. (b). Boo wanted something.
   i. a) entails (b), if what Boo wanted was the crown, then Boo wanted something at least. Hence, if (a) is **TRUE**, (b) must also be **TRUE**.
   ii. NOT-(a) entails (b), if it is not the case that what Boo wanted was the crown, then Boo still wanted something. Hence, if NOT-(a) is **TRUE**, (b) must also be **TRUE**.
   iii. Since both (a) and NOT-(a) entail (b), (a) **PRESUPPOSES** (b)

5. (a). Daisy finished reading the paper. (b). Daisy started reading the paper.

   i. a) entails (b), if Daisy finished reading the paper, then she must have started reading. Hence, if (a) is **TRUE**, (b) must also be **TRUE**.

   ii. NOT-(a) does not entail (b), if Daisy did not finish reading the paper, it is not necessarily true she has started reading, it can be the case that she has not started at all. Hence, if NOT-(a) is **TRUE**, (b) is not necessarily **TRUE**.

   iii. (a) entails (b), but NOT-(a) does not. Hence, (a) **does not PRESUPPOSE** (b)

Part 2. Set Theory ***[difficulty level: ☆☆]***

Answer the following questions, assuming the following subsets.

1. **State whether the expression is true or false** and **briefly explain why**.

a. $|M| = |H|$

true, |M| = |H | = 7

the cardinality of the set M and the set H are both 7

b. $G \cap T \neq \emptyset$

true, G ∩ T = { Yoshi, Bowser, Petey Piranha } ≠ ∅

this set means "characters who wear green and whose tongue is visible", and we have at least Yoshi be such a character, so it is not an empty set

c. King Boo ∈ ($N \cap H$)

false, N ∩ H = { Yoshi, Bowser } and King Boo is not in this set

this set means characters who are humanoid and who have a big nose simultaneously, and King Boo does not have big nose

d. $T \subseteq M$

false, for T ⊆ M, every member of T should be in M as well, but Mario is not

this means that everyone whose tongue is visible has a giant mouth at the same time, but Mario does not

e. Peach ∈ C'

true, C' = { Peach, Yoshi, Toad, Daisy, Birdo, Toadette, Bowser, Petey Piranha, King Boo } and Peach ∈ C'

this means that peach is not a character that wears a cap, which is true

f. $K - M$ = { King Boo }

false, K – M = { Peach, Daisy }

this set means characters who are crowned but do not have a giant mouth, and King Boo does have a giant mouth

2. **Calculate** the following and **describe the resulting set in words**.

a. $N \cap H$

calculated set: { Yoshi, Bowser }

description: humanoid characters that have a big nose

b. $G - C$

calculated set: { Peach, Yoshi, Daisy, Bowser, Petey Piranha }

description: characters that wear green but not a cap

c. $K \cap H'$

calculated set: { Peach, Daisy }

description: non-humanoid characters that are crowned

d. $(K \cup N)'$

calculated set: { Toad, Birdo, Petey Piranha, Toadette }

description: uncrowned-characters who do not have big noses

e. $T - (K \cup M)$

calculated set: { Mario }

description: uncrowned characters whose tongue is visible but do not have giant mouths

f. $(H - M) \cup (K \cap T)$

calculated set: { Toad, Toadette, King Boo }

description: characters who are either (humanoid without a giant mouth) OR (crowned with visible tongue)

3. Translate the following sentences into set-theoretic notation by following semantics of quantifiers.

   a. Some characters with giant mouths have big noses.

      $M \cap N \neq \emptyset$, or $|M \cap N| \neq 0$

   b. No crowned characters wear a cap.

      $K \cap C = \emptyset$, or $|K \cap C| = 0$